Jarish Ahamad

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	0.00
Jawahar Navodaya Vidyalaya	Class XII (CBSE)	2009	83.00%
Jawahar Navodaya Vidyalaya	Class X (CBSE)	2007	82.60%

EDUCATION / ACADEMIC QUALIFICATIONS

PhD in Organic Chemistry -- Department of Chemistry, Indian Institute of Technology Hyderabad (2021-Present). Expected OC (Open Colloquium): Oct 2026. PhD coursework CGPA: 9.0/10.0.

M.Sc. in Chemistry -- Department of Chemistry, Indian Institute of Technology Delhi. Percentage: 70.0%.

B.Sc. (Hons.) in Chemistry -- Ramjas College, University of Delhi. Percentage: 61.0%.

Class XII (Science) -- Jawahar Navodaya Vidyalaya--Percentage: 83.0%.

Class X (Science) -- Jawahar Navodaya Vidyalaya--Percentage: 82.6%.

SCHOLASTIC ACHIEVEMENTS

- Senior Research Fellowship (SRF), CSIR, India, 2023.
- Junior Research Fellowship (JRF), CSIR, India, 2021.
- CSIR-UGC NET-JRF qualified, 2020.
- GATE (Chemistry) qualified, 2020.
- IIT-JAM (M.Sc. Chemistry) qualified, 2012.

PUBLICATIONS

1. Ahamad, J.; Khan, F. A. Stereochemical Revision and First Total Synthesis of Renifolin D. *J. Nat. Prod.* **2026**, *89*, 1524-1531. https://doi.org/10.1021/acs.jnatprod.6c00156

2. Ahamad, J.; Khan, F. A. Biomimetic Total Syntheses of Renifolin F and Antiarone K. *Org. Biomol. Chem.* **2024**, *22* (24), 4877-4881. https://doi.org/10.1039/D4OB00651H

3. Ahamad, J.; Khatua, R. R.; Khan, F. A. Lewis Acid-Promoted Intramolecular Cyclization of *Ortho* -Prenylated Chalcones: Application to First Total Synthesis of (±) Involucrasin C and Its Novel Analogues. *RSC Adv.* **2025**, *15* (33), 27254-27259. https://doi.org/10.1039/D5RA04830C

4. Ahamad, J.; Khan, F. A. *Eur. J. Org. Chem*. **2026** Diastereoselective Synthesis of Indane Scaffolds via Acid-Mediated Cascade Cyclisation (revision submitted).

5. Ahamad, J.; Khan, F. A. Diastereoselective and Regioselective Construction of Acyloins and Oxazolines via Cascade Double Annulation (to be submitted).

6. Rahul, N.; **Ahamad, J**.; Khan, F. A. Total Synthesis of (±) Involucrasin A, B and Antiarone M (to be submitted).

RESEARCH EXPERIENCE

Ph.D. Research -- Synthetic Organic Chemistry

Department of Chemistry, Indian Institute of Technology Hyderabad

Supervisor: Prof. Faiz Ahmed Khan | 2021-Present

Conducting research on the total synthesis of prenylated natural products, stereochemical revision, and prenyl group chemistry.

Developing synthetic methodologies and cascade cyclisation reactions for the construction of complex organic scaffolds.

Performing multistep synthesis, reaction optimisation, purification, and structural characterisation of organic compounds.

Investigating reaction pathways, stereochemical outcomes, and structure-reactivity relationships.

Applying spectroscopic and analytical techniques to establish molecular structures and confirm product identity.

PROJECTS

Project Experience

Title: "Functionalised material for water purification"

Worked on the characterisation of rice husk and rice husk ash and explored their application as low-cost adsorbents for water purification. Additionally, performed surface modification of multi-walled carbon nanotubes (MWCNTs) to enhance their efficiency for the removal of toxic ions from contaminated water. Evaluated material properties and adsorption performance for environmental remediation applications (Master's project).

Supervisor: Prof. Ashok Kumar Ganguli (IIT Delhi)

TECHNICAL SKILLS

- Multi-step Synthesis and total Synthesis of natural products
- Synthetic Methodology Development, Reaction Optimisation, and Scale-up Synthesis
- Air- and Moisture-Sensitive Synthesis (Schlenk Techniques)
- Heterocyclic and Medicinally Relevant Molecule Synthesis
- Organometallic Chemistry and Transition Metal-Catalysed Cross-Coupling Reactions
- Purification Techniques (Column Chromatography, Flash Chromatography, Recrystallisation)
- Structure Elucidation and Spectral Interpretation
- Scientific Literature Search

ANALYTICAL & CHARACTERIZATION SKILLS

- NMR Spectroscopy (^{1}H, ^{13}C, DEPT, and 2D NMR)
- Mass Spectrometry
- IR Spectroscopy
- Single-Crystal X-ray Crystallography

SOFTWARE SKILLS

- ChemDraw
- MestReNova
- SciFinder
- Zotero
- Mendeley
- Olex2
- Microsoft Office

CONFERENCES ATTENDED

- Attended the 23rd Lecture Conference on Organic Chemistry (ORCHEM-2024) at University of Regensburg, Germany, September 9-11, 2024.
- Attended the XX J-National Organic Symposium Trust Conference on Organic Chemistry (XX J-NOST-2025), at IIT (ISM) Dhanbad, India, 29 Oct-01Nov, 2025.
- Attended the International Chemistry Conference on Current & Future Trends in Chemical Sciences (Chem Spirit-2025), at IIT Hyderabad, India, November 17-19, 2025.

POSITIONS OF RESPONSIBILITY

Teaching Assistant (TA), Department of Chemistry, IIT Hyderabad

Assisted in conducting chemistry laboratory sessions and guiding M.Sc. students during practical work.

Research Support for M.Sc. Project Students, Department of Chemistry, IIT Hyderabad

Provided experimental guidance and research support to M.Sc. students undertaking their project work in the laboratory.

PROFESSIONAL PROFILE

I am Jarish Ahamad, a PhD candidate in Organic Chemistry. My research expertise lies in synthetic organic chemistry, with extensive experience in multi-step synthesis, synthetic methodology development, reaction optimisation, and structure elucidation. I have hands-on experience in the design and synthesis of complex organic molecules, purification, and characterisation using modern spectroscopic techniques. I am proficient in conducting independent research and collaborating effectively within multidisciplinary teams. With strong analytical and problem-solving skills, I am keen to contribute to the design and synthesis of novel small molecules, NCE discovery, and pharmaceutical research and development.

RELEVANT COURSES

- Chemistry of Natural Product Biomolecules
- Drug Discovery, Design and Development
- Spectroscopy & Applications